

Overview

- Overview of Cryptography
- Classical Symmetric Cipher
 - Substitution Cipher
 - Transposition Cipher
 - Product Cipher
- Modern Symmetric Ciphers (DES)

Basic Terminology

- **plaintext** - the original message
- **ciphertext** - the coded message
- **cipher** - algorithm for transforming plaintext to ciphertext
- **key** - info used in cipher known only to sender/receiver
- **encipher (encrypt)** - converting plaintext to ciphertext
- **decipher (decrypt)** - recovering ciphertext from plaintext
- **cryptography** - study of encryption principles/methods
- **cryptanalysis (codebreaking)** - the study of principles/methods of deciphering ciphertext *without* knowing key

Classification of Cryptography

- Number of keys used
 - Hash functions: no key
 - Secret key cryptography: one key
 - Public key cryptography: two keys - public, private
- Type of encryption operations used
 - substitution / transposition / product
- Way in which plaintext is processed
 - block / stream

Secret Key vs. Secret Algorithm

- Secret algorithm: additional hurdle
- Hard to keep secret if used widely:
 - Reverse engineering, social engineering
- Commercial: published
 - Wide review, trust
- Military: avoid giving enemy good ideas

Cryptanalysis Scheme

- Ciphertext only:
 - Exhaustive search until "recognizable plaintext"
 - Need enough ciphertext
- Known plaintext:
 - Secret may be revealed (by spy, time), thus <ciphertext, plaintext> pair is obtained
 - Great for monoalphabetic ciphers
- Chosen plaintext:
 - Choose text, get encrypted
 - Useful if limited set of messages

Unconditional vs. Computational Security

- **Unconditional security**

- No matter how much computer power is available, the cipher cannot be broken
- The ciphertext provides insufficient information to uniquely determine the corresponding plaintext
- Only one-time pad scheme qualifies

- **Computational security**

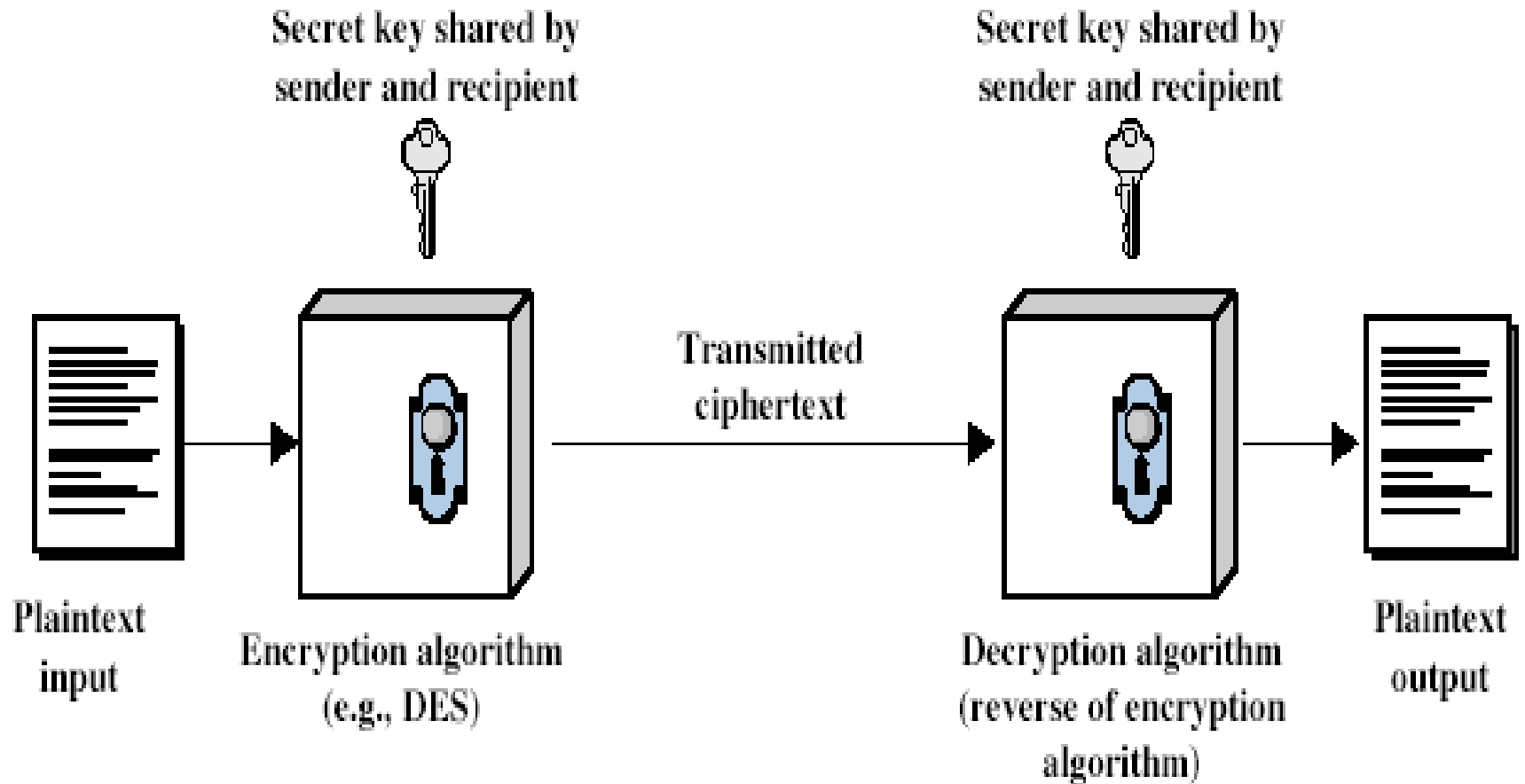
- The cost of breaking the cipher exceeds the value of the encrypted info
- The time required to break the cipher exceeds the useful lifetime of the info

Brute Force Search

- Always possible to simply try every key
- Most basic attack, proportional to key size
- Assume either know / recognise plaintext

Key Size (bits)	Number of Alternative Keys	Time required at 1 encryption/ μ s	Time required at 10^6 encryptions/ μ s
32	$2^{32} = 4.3 \times 10^9$	$2^{31} \mu\text{s} = 35.8 \text{ minutes}$	2.15 milliseconds
56	$2^{56} = 7.2 \times 10^{16}$	$2^{55} \mu\text{s} = 1142 \text{ years}$	10.01 hours
128	$2^{128} = 3.4 \times 10^{38}$	$2^{127} \mu\text{s} = 5.4 \times 10^{24} \text{ years}$	$5.4 \times 10^{18} \text{ years}$
168	$2^{168} = 3.7 \times 10^{50}$	$2^{167} \mu\text{s} = 5.9 \times 10^{36} \text{ years}$	$5.9 \times 10^{30} \text{ years}$
26 characters (permutation)	$26! = 4 \times 10^{26}$	$2 \times 10^{26} \mu\text{s} = 6.4 \times 10^{12} \text{ years}$	$6.4 \times 10^6 \text{ years}$

Symmetric Cipher Model



Requirements

- Two requirements for secure use of symmetric encryption:
 - a strong encryption algorithm
 - a secret key known only to sender / receiver
- $$Y = E_k(X)$$
- $$X = D_k(Y)$$
- Assume encryption algorithm is known
 - Implies a secure channel to distribute key

Classical Substitution Ciphers

- Letters of plaintext are replaced by other letters or by numbers or symbols
- Plaintext is viewed as a sequence of bits, then substitution replaces plaintext bit patterns with ciphertext bit patterns

Caesar Cipher

- Earliest known substitution cipher
- Replaces each letter by 3rd letter on
- Example:

meet me after the toga party

PHHW PH DIWHU WKH WRJD SDUWB

Caesar Cipher

- Define transformation as:

a b c d e f g h i j k l m n o p q r s t u v w x y z

D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

- Mathematically give each letter a number

a b c d e f g h i j k l m

0 1 2 3 4 5 6 7 8 9 10 11 12

n o p q r s t u v w x y z

13 14 15 16 17 18 19 20 21 22 23 24 25

- Then have Caesar cipher as:

$$C = E(p) = (p + k) \bmod (26)$$

$$p = D(C) = (C - k) \bmod (26)$$

Cryptanalysis of Caesar Cipher

- Only have 25 possible ciphers
 - A maps to B,..Z
- Given ciphertext, just try all shifts of letters
- Do need to recognize when have plaintext
- E.g., break ciphertext "GCUA VQ DTGCM"

Monoalphabetic Cipher

- Rather than just shifting the alphabet
- Could shuffle (jumble) the letters arbitrarily
- Each plaintext letter maps to a different random ciphertext letter
- Key is 26 letters long

Plain: abcdefghijklmnopqrstuvwxyz

Cipher: DKVQFIBJWPESCXHTMYAUOLRGZN

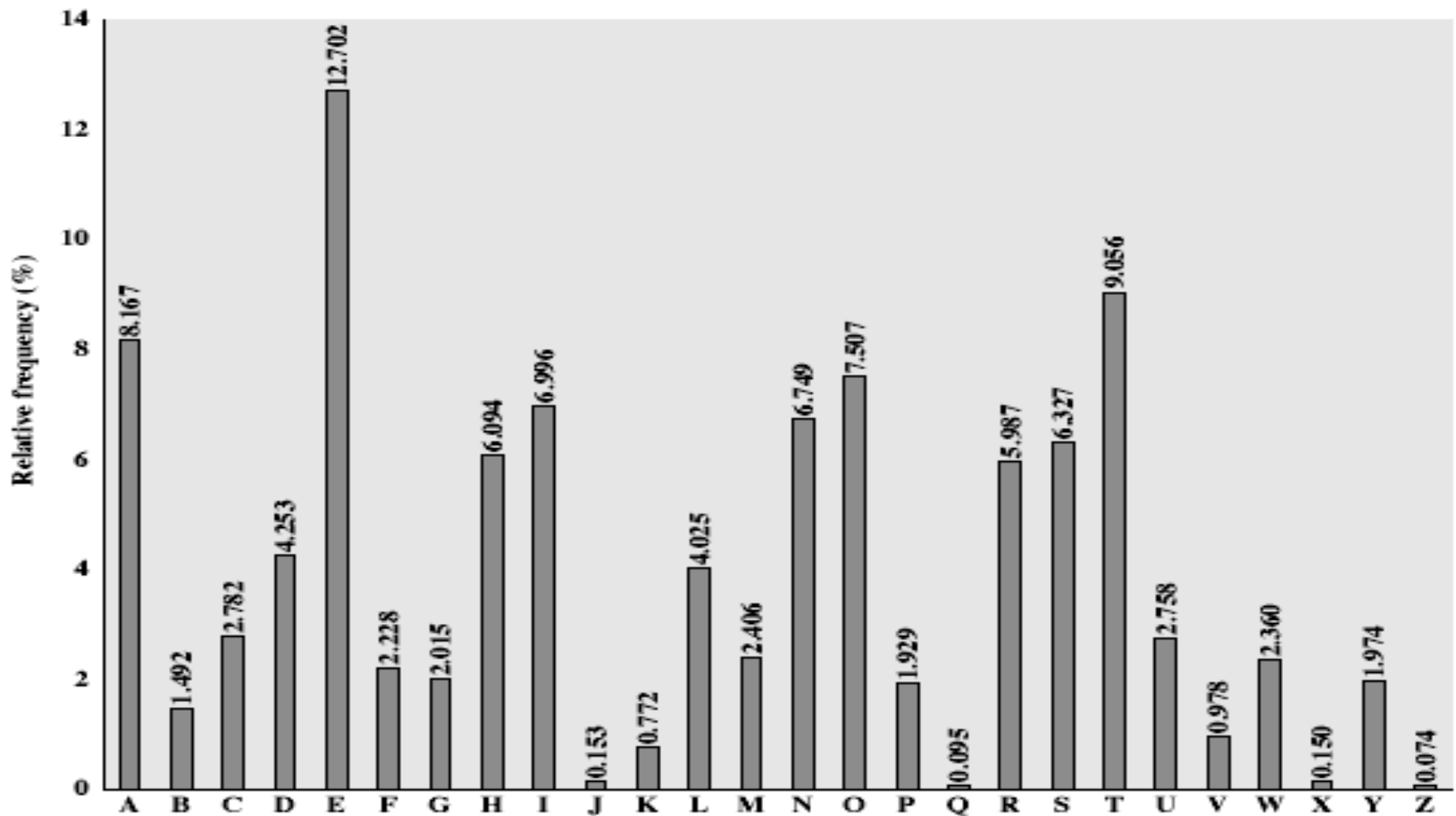
Plaintext: ifwewishtoreplaceletters

Ciphertext: WIRFRWAJUHYFTSDVFSFUUFYA

Monoalphabetic Cipher Security

- Now have a total of $26! = 4 \times 10^{26}$ keys
- Is that secure?
- Problem is language characteristics
 - Human languages are **redundant**
 - Letters are not equally commonly used

English Letter Frequencies



Example Cryptanalysis

- Given ciphertext:

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ

VUEPHZHMZSHZOWSFPAPPDTSVPQUZWYMXUZHUSX

EPYEPOPDZSZUFPOMBZWPFPUPZHMDJUDTMOHMQ

- Count relative letter frequencies (see text)
- Guess P & Z are e and t
- Guess ZW is th and hence ZWP is the
- Proceeding with trial and error finally get:

it was disclosed yesterday that several informal but
direct contacts have been made with political
representatives of the viet cong in moscow

One-Time Pad

- If a truly random key as long as the message is used, the cipher will be secure - One-Time pad
- E.g., a random sequence of 0's and 1's XORed to plaintext, no repetition of keys
- Unbreakable since ciphertext bears no statistical relationship to the plaintext
- For **any** plaintext, it needs a random key of the same length
 - Hard to generate large amount of keys
- Have problem of safe distribution of key

Transposition Ciphers

- Now consider classical **transposition** or **permutation** ciphers
- These hide the message by rearranging the letter order, without altering the actual letters used
- Can recognise these since have the same frequency distribution as the original text

Rail Fence cipher

- Write message letters out diagonally over a number of rows
- Then read off cipher row by row
- E.g., write message out as:

m e m a t r h t g p r y

e t e f e t e o a a t

- Giving ciphertext

MEMATRHTGPRYETEFETEOAAT

Product Ciphers

- Ciphers using substitutions or transpositions are not secure because of language characteristics
- Hence consider using several ciphers in succession to make harder, but:
 - Two substitutions make a more complex substitution
 - Two transpositions make more complex transposition
 - But a substitution followed by a transposition makes a new much harder cipher
- This is bridge from classical to modern ciphers

Block vs Stream Ciphers

- Block ciphers process messages in into blocks, each of which is then en/decrypted
- Like a substitution on very big characters
 - 64-bits or more
- Stream ciphers process messages a bit or byte at a time when en/decrypting
- Many current ciphers are block ciphers, one of the most widely used types of cryptographic algorithms

Block Cipher Principles

- Most symmetric block ciphers are based on a **Feistel Cipher Structure**
- Block ciphers look like an extremely large substitution
- Would need table of 2^{64} entries for a 64-bit block
- Instead create from smaller building blocks
- Using idea of a product cipher

Substitution-Permutation Ciphers

- Substitution-permutation (S-P) networks
[Shannon, 1949]
 - modern substitution-transposition product cipher
- These form the basis of modern block ciphers
- S-P networks are based on the two primitive cryptographic operations
 - *substitution* (S-box)
 - *permutation* (P-box)
- provide *confusion* and *diffusion* of message

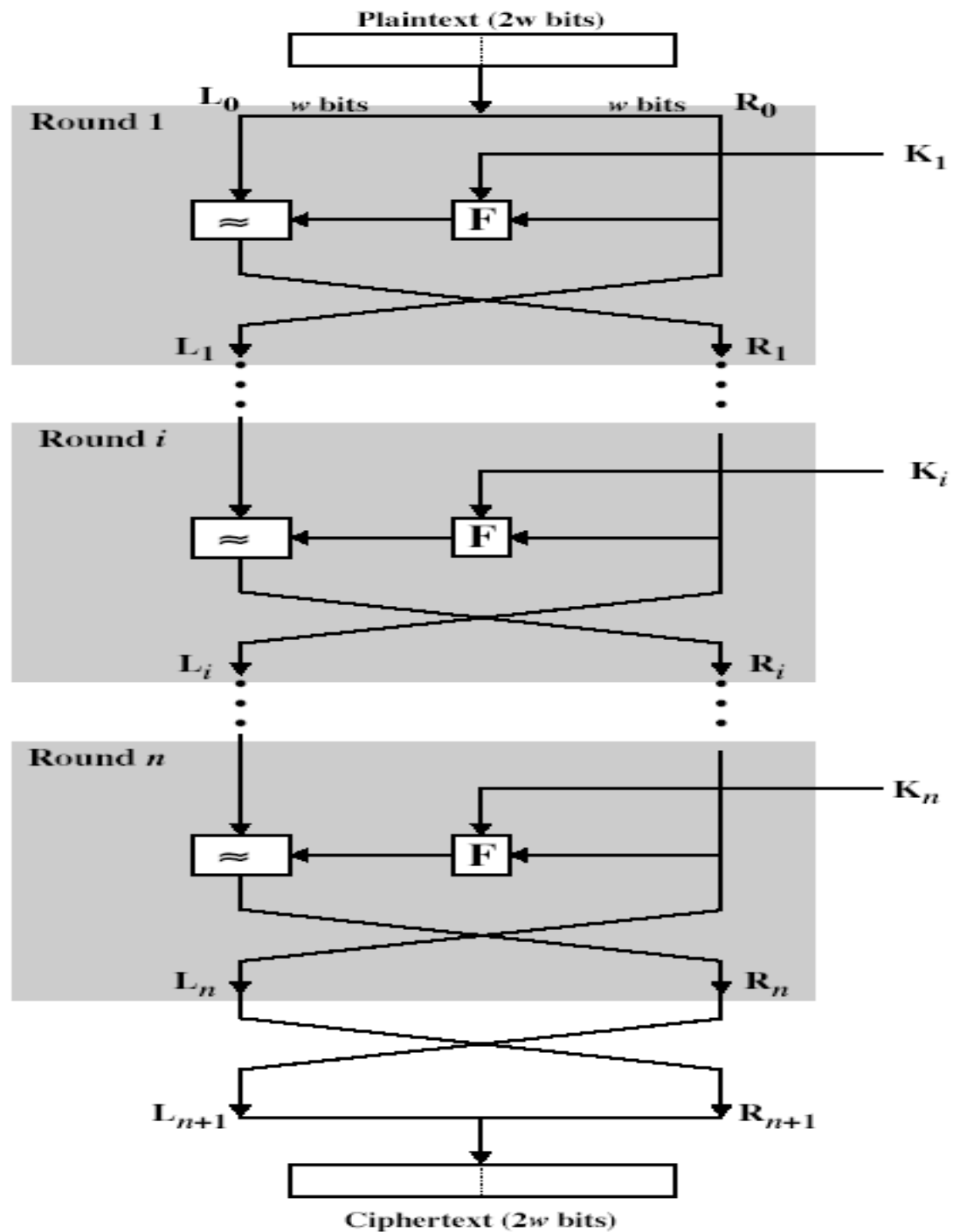
Confusion and Diffusion

- Cipher needs to completely obscure statistical properties of original message
- A one-time pad does this
- More practically Shannon suggested S-P networks to obtain:
- **Diffusion** - dissipates statistical structure of plaintext over bulk of ciphertext
- **Confusion** - makes relationship between ciphertext and key as complex as possible

Feistel Cipher Structure

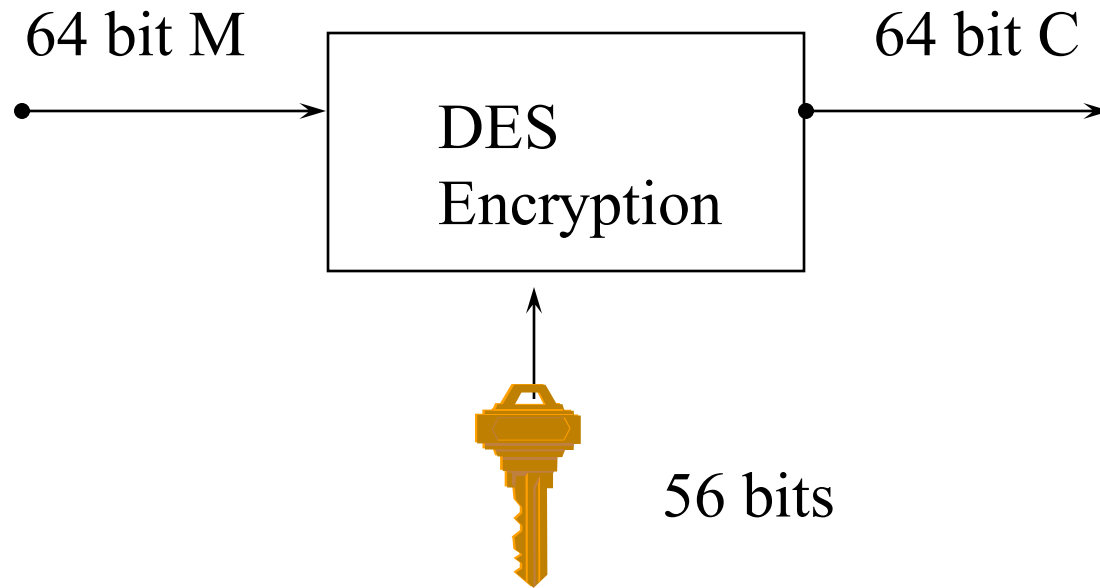
- **Feistel cipher** implements Shannon's S-P network concept
 - based on invertible product cipher
- Process through multiple rounds which
 - partitions input block into two halves
 - perform a substitution on left data half
 - based on round function of right half & subkey
 - then have permutation swapping halves

Feistel Cipher Structure

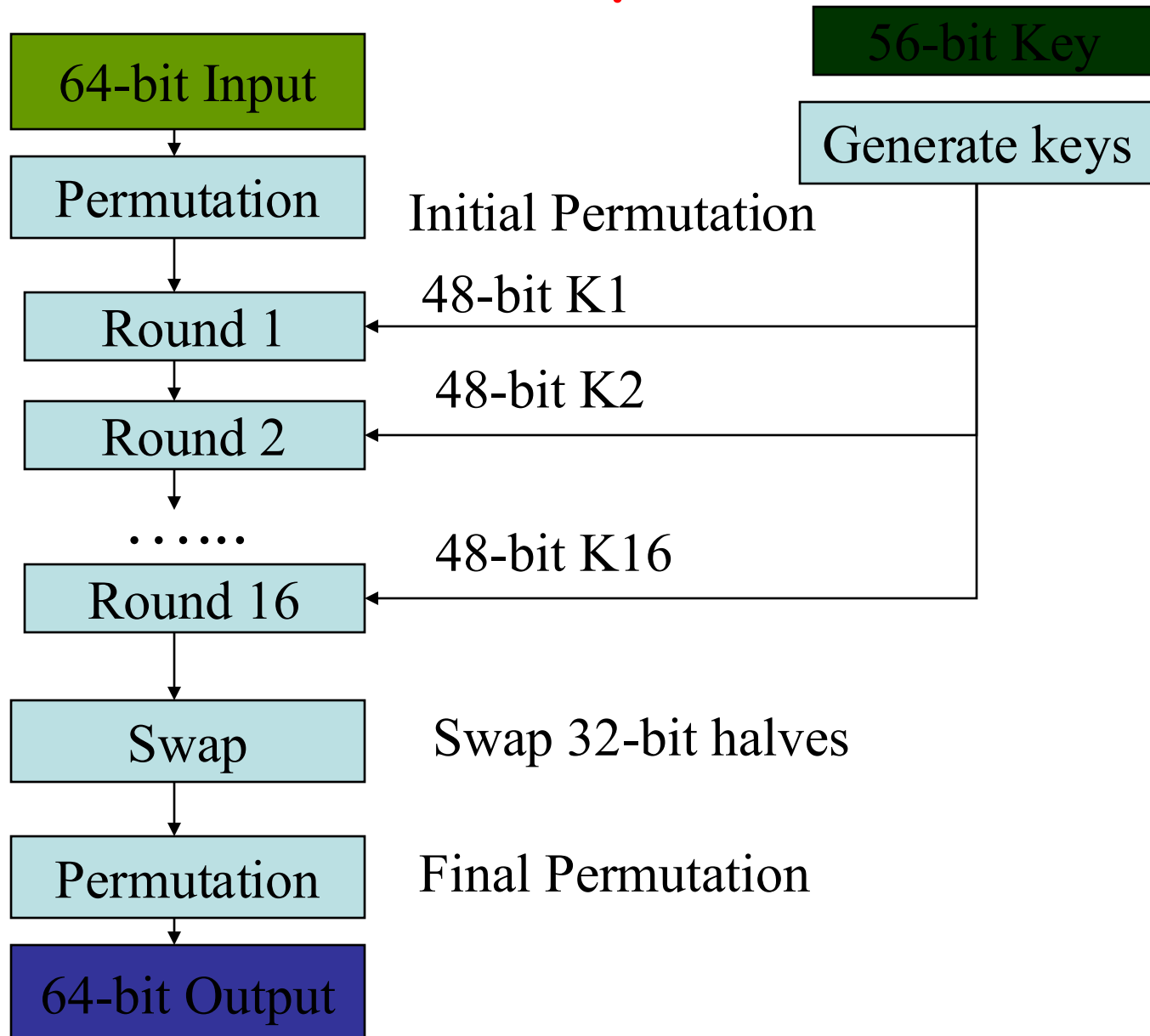


DES (Data Encryption Standard)

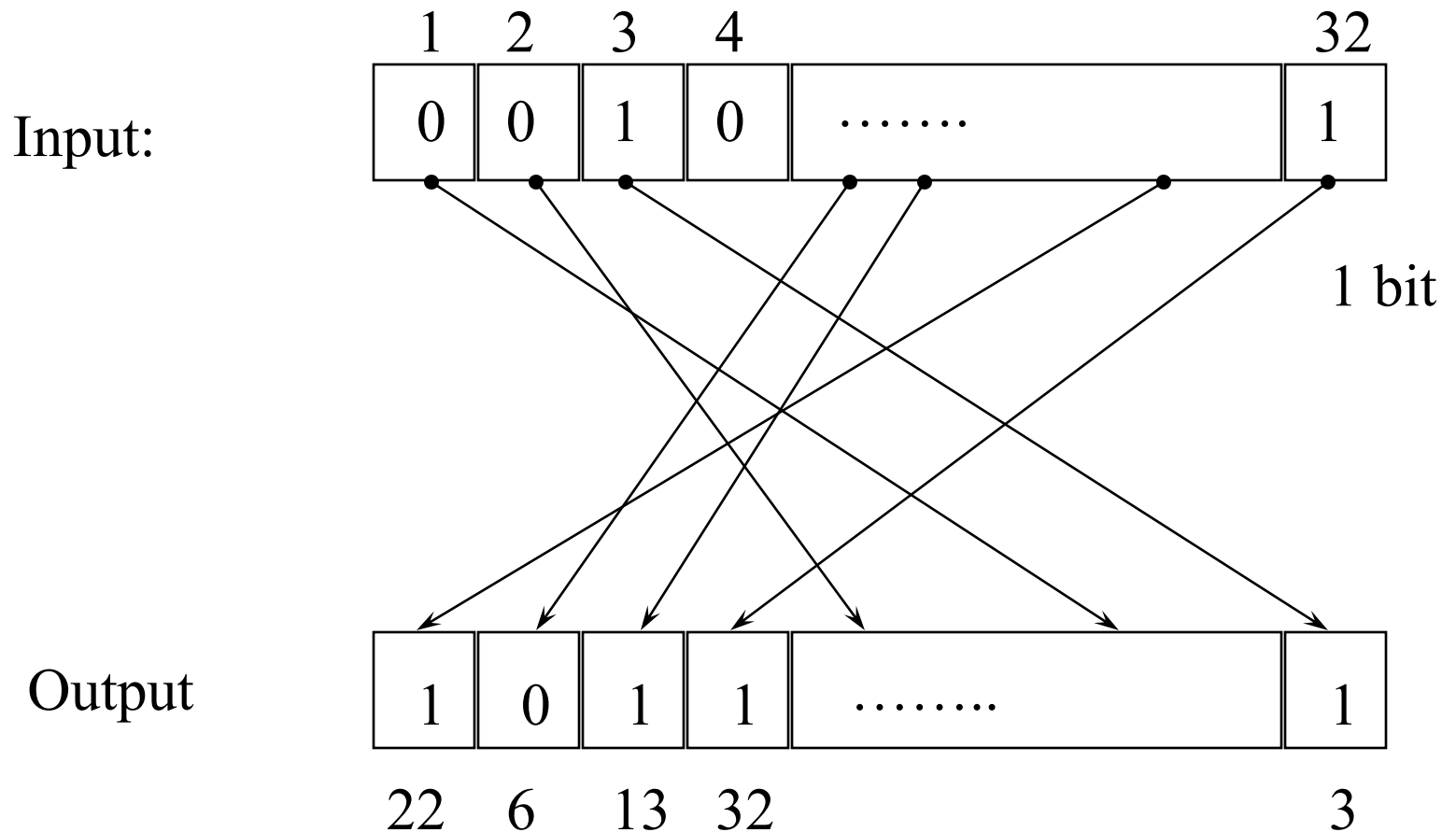
- Published in 1977, standardized in 1979.
- Key: 64 bit quantity=8-bit parity+56-bit key
 - Every 8th bit is a parity bit.
- 64 bit input, 64 bit output.



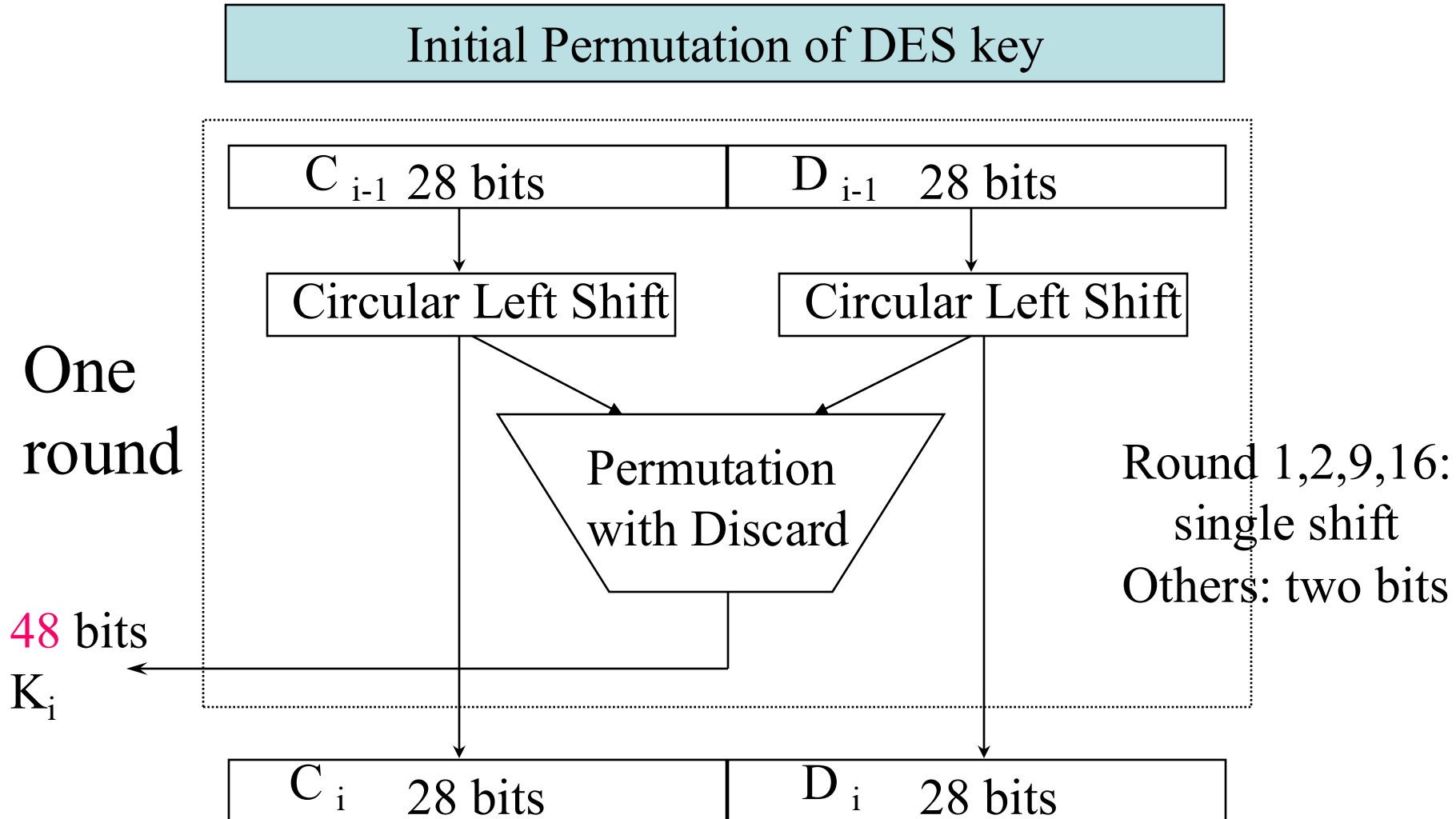
DES Top View



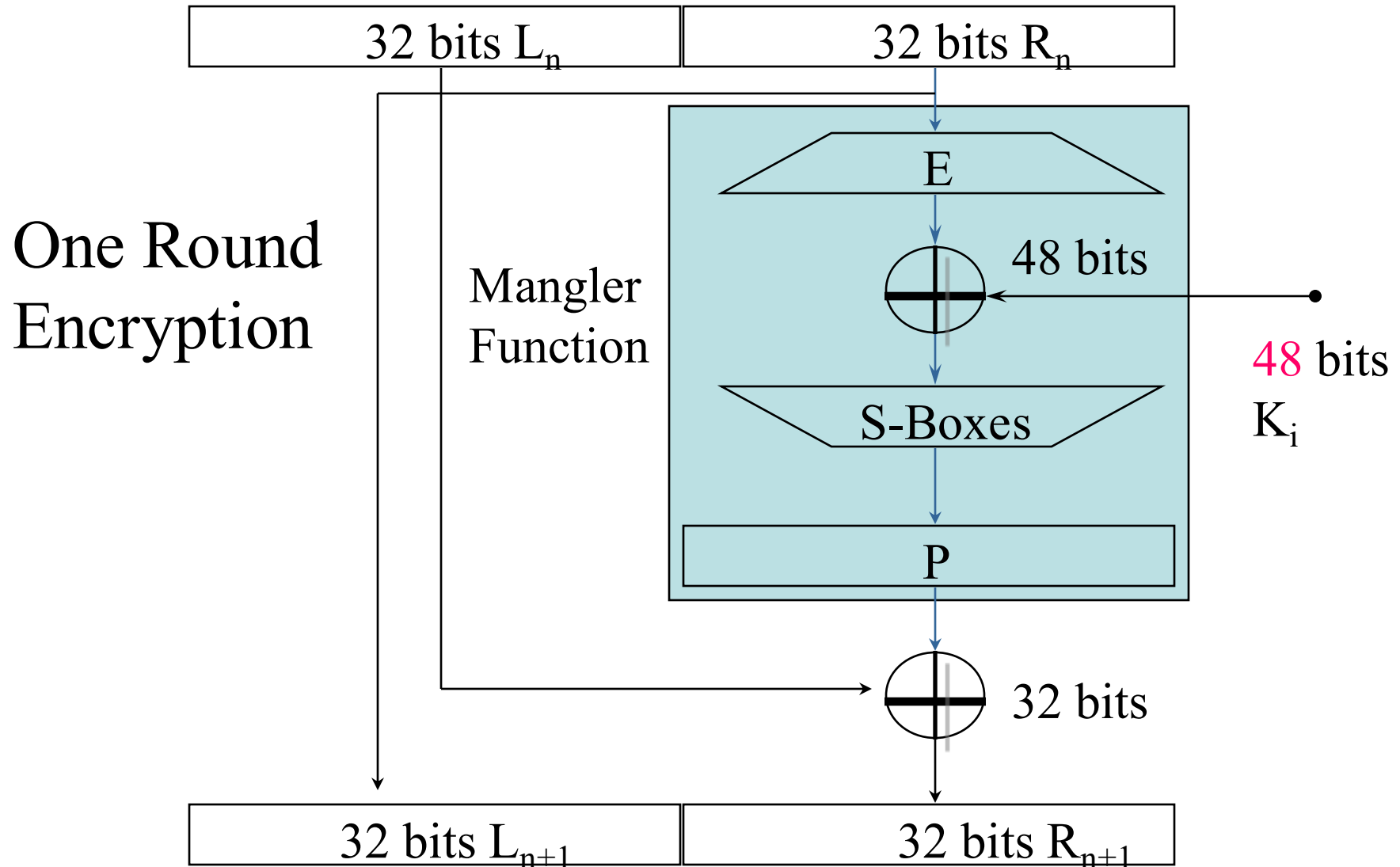
Bit Permutation (1-to-1)



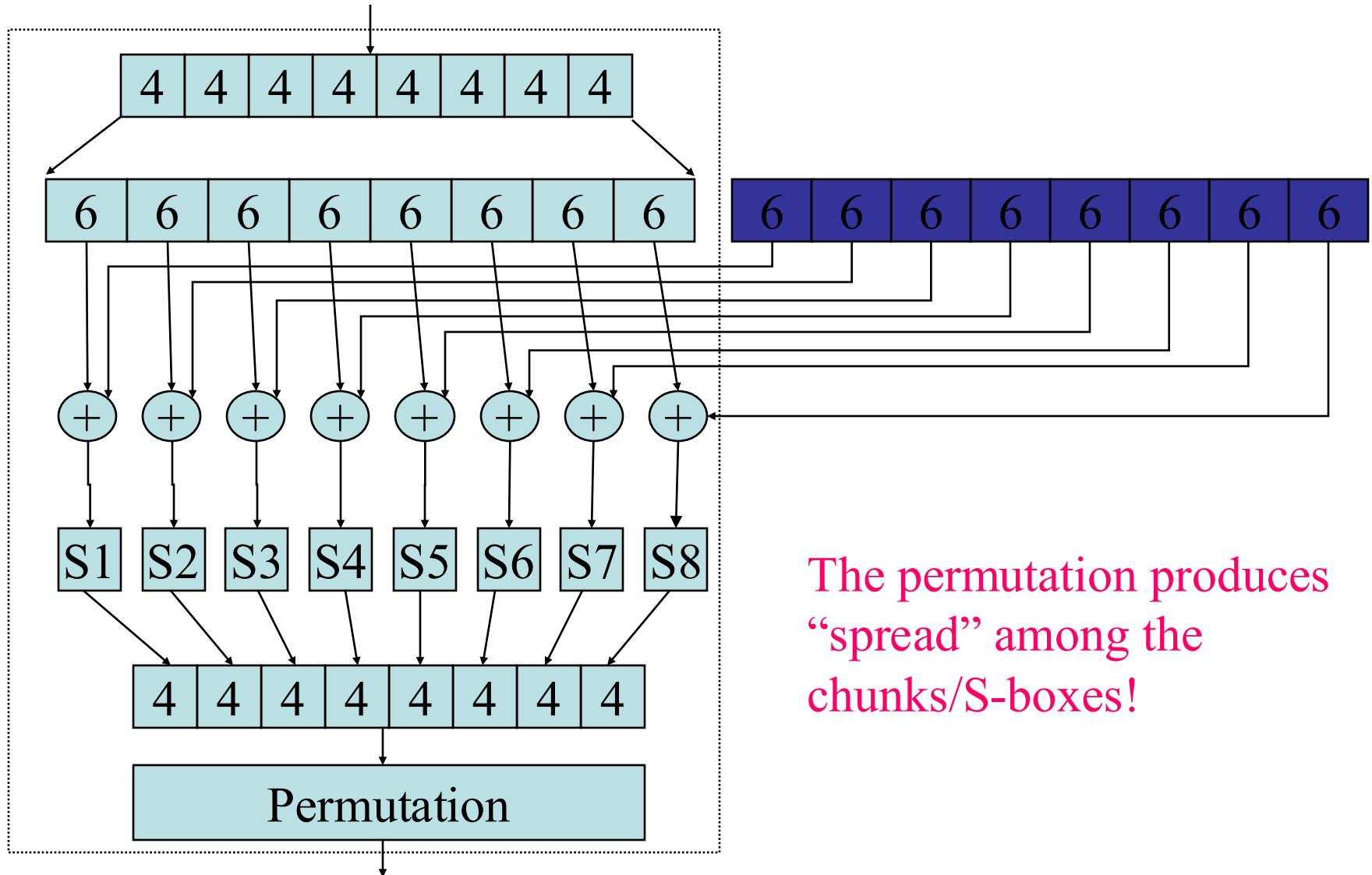
Per-Round Key Generation



A DES Round

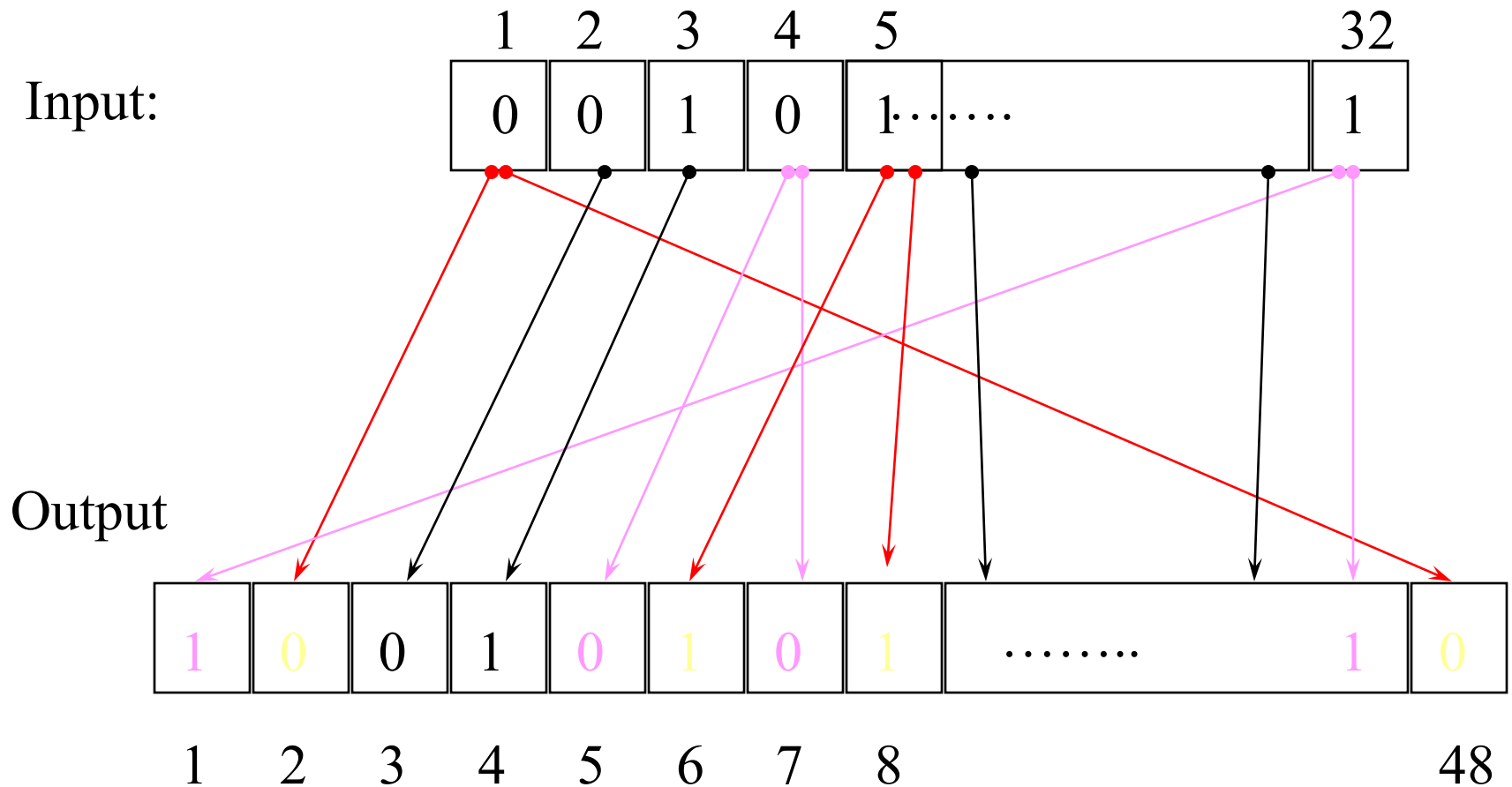


Mangler Function



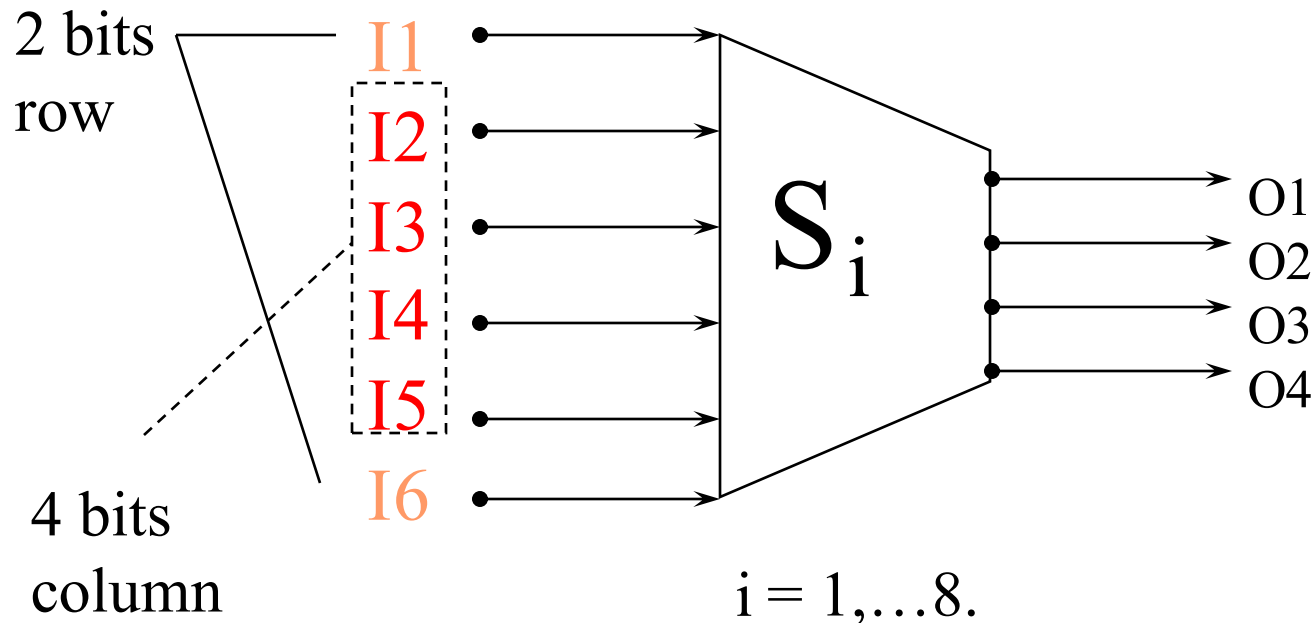
The permutation produces
“spread” among the
chunks/S-boxes!

Bits Expansion (1-to-m)



S-Box (Substitute and Shrink)

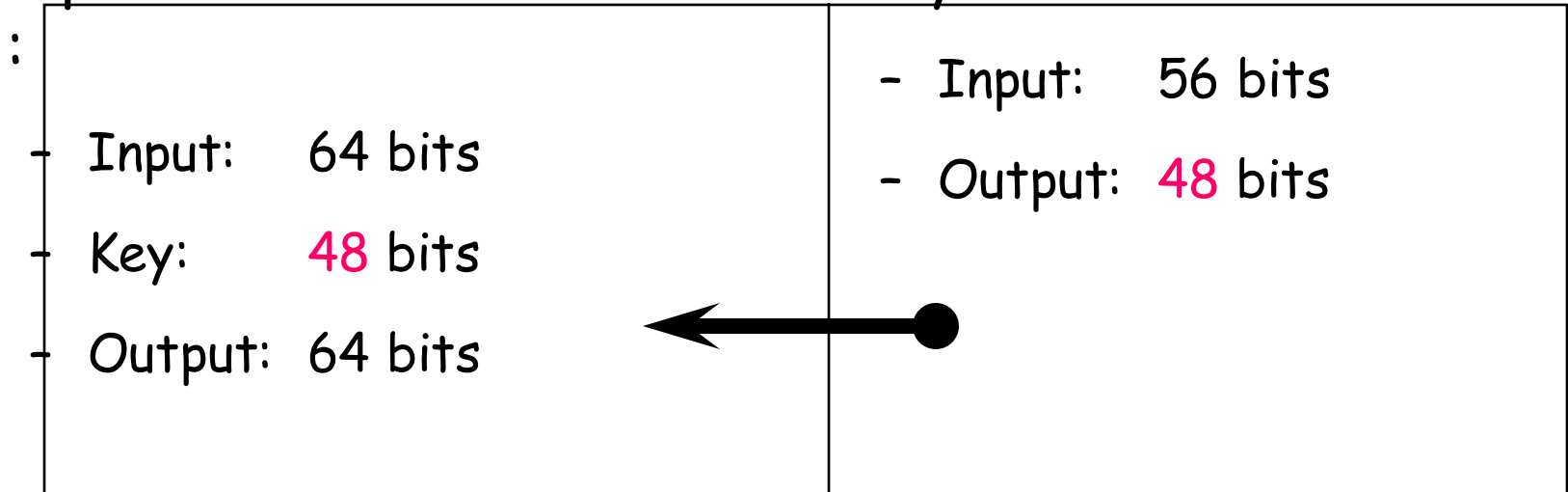
- 48 bits $\Rightarrow$ 32 bits. ($8 \times 6 \Rightarrow 8 \times 4$)
- 2 bits used to select amongst 4 substitutions for the rest of the 4-bit quantity



DES Standard

- Cipher Iterative Action

- Key Generation Box :



One round (Total 16 rounds)

DES Box Summary

- Simple, easy to implement:
 - Hardware/gigabits/second,
software/megabits/second
- 56-bit key DES may be acceptable for non-critical applications but triple DES (DES3) should be secure for most applications today
- Supports several operation modes (ECB CBC, OFB, CFB) for different applications

Outlines

- Strength/weakness of DES, AES
- Public Key Cryptography
- Modular Arithmetic
- RSA

Avalanche Effect

- Key desirable property of encryption alg
- Where a change of **one** input or key bit results in changing more than **half** output bits
- DES exhibits strong avalanche

Strength of DES - Key Size

- 56-bit keys have $2^{56} = 7.2 \times 10^{16}$ values
- Brute force search looks hard
- Recent advances have shown is possible
 - in 1997 on a huge cluster of computers over the Internet in a few months
 - in 1998 on dedicated hardware called "DES cracker" by EFF in a few days (\$220,000)
 - in 1999 above combined in 22hrs!
- Still must be able to recognize plaintext
- No big flaw for DES algorithms

DES Replacement

- Triple-DES (3DES)
 - 168-bit key, no brute force attacks
 - Underlying encryption algorithm the same, no effective analytic attacks
 - Drawbacks
 - Performance: no efficient software codes for DES/3DES
 - Efficiency/security: bigger block size desirable
- Advanced Encryption Standards (AES)
 - US NIST issued call for ciphers in 1997
 - Rijndael was selected as the AES in Oct-2000

AES

- Private key symmetric block cipher
- 128-bit data, 128/192/256-bit keys
- Stronger & faster than Triple-DES
- Provide full specification & design details
- Evaluation criteria
 - Security: effort to practically cryptanalysis
 - Cost: computational efficiency and memory requirement
 - Algorithm & implementation characteristics: flexibility to apps, hardware/software suitability, simplicity

AES Shortlist

- After testing and evaluation, shortlist in Aug-99:
 - MARS (IBM) - complex, fast, high security margin
 - RC6 (USA) - v. simple, v. fast, low security margin
 - Rijndael (Belgium) - clean, fast, good security margin
 - Serpent (Euro) - slow, clean, v. high security margin
 - Twofish (USA) - complex, v. fast, high security margin
- Then subject to further analysis & comment

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Private-Key Cryptography

- Private/secret/single key cryptography uses one key
- Shared by both sender and receiver
- If this key is disclosed communications are compromised
- Also is **symmetric**, parties are equal
- Hence does not protect sender from receiver forging a message & claiming is sent by sender

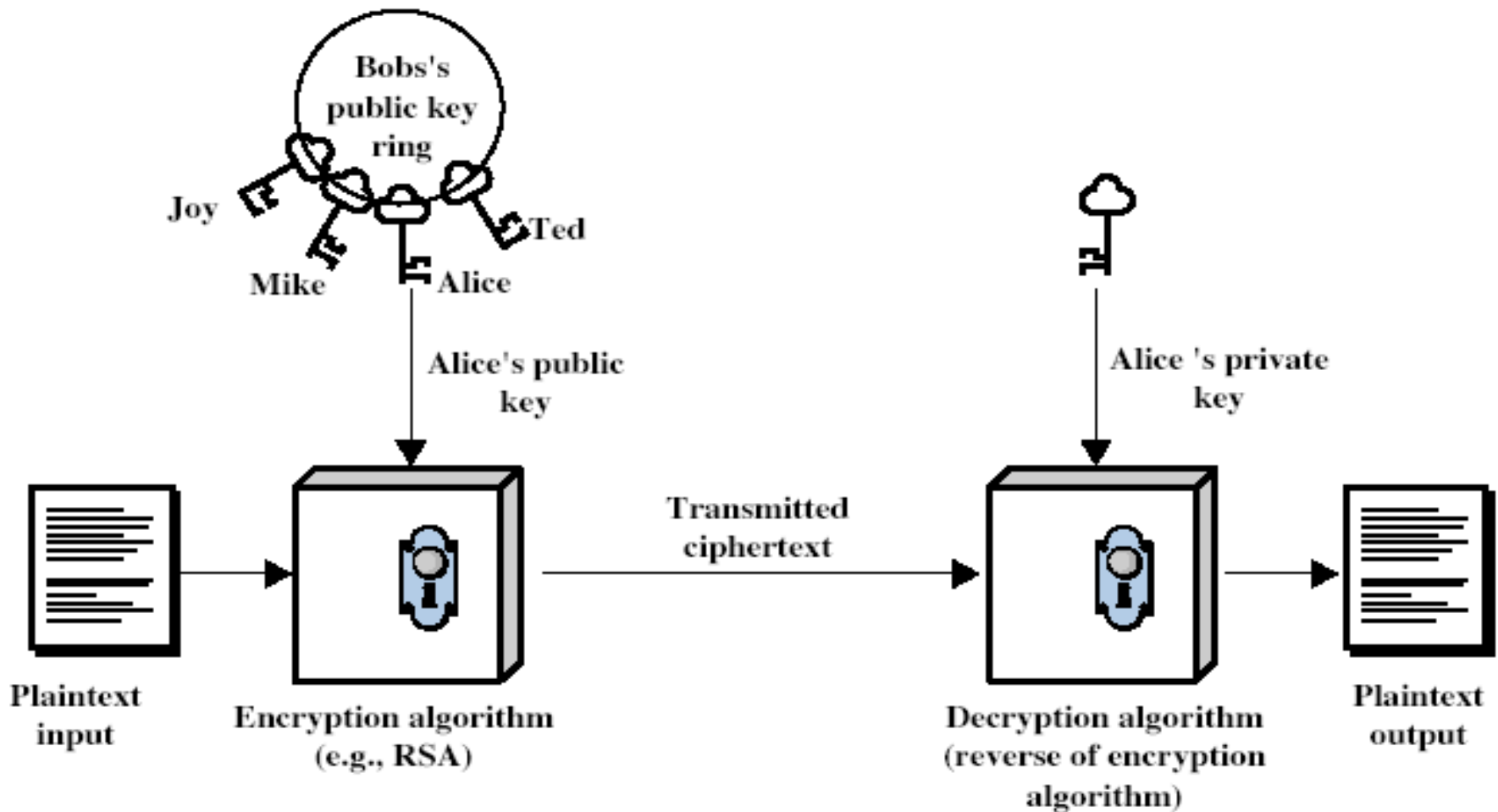
Public-Key Cryptography

- Probably most significant advance in the 3000 year history of cryptography
- Uses **two** keys - a public & a private key
- **Asymmetric** since parties are **not** equal
- Uses clever application of number theoretic concepts to function
- Complements **rather than** replaces private key crypto

Public-Key Cryptography

- **Public-key/two-key/asymmetric** cryptography involves the use of **two** keys:
 - a **public-key**, which may be known by anybody, and can be used to **encrypt messages**, and **verify signatures**
 - a **private-key**, known only to the recipient, used to **decrypt messages**, and **sign (create) signatures**
- **Asymmetric** because
 - those who encrypt messages or verify signatures **cannot** decrypt messages or create signatures

Public-Key Cryptography



Public-Key Characteristics

- Public-Key algorithms rely on two keys with the characteristics that it is:
 - computationally infeasible to find decryption key knowing only algorithm & encryption key
 - computationally easy to en/decrypt messages when the relevant (en/decrypt) key is known
 - either of the two related keys can be used for encryption, with the other used for decryption (in some schemes)
- Analogy to delivery w/ a padlocked box

Public-Key Cryptosystems

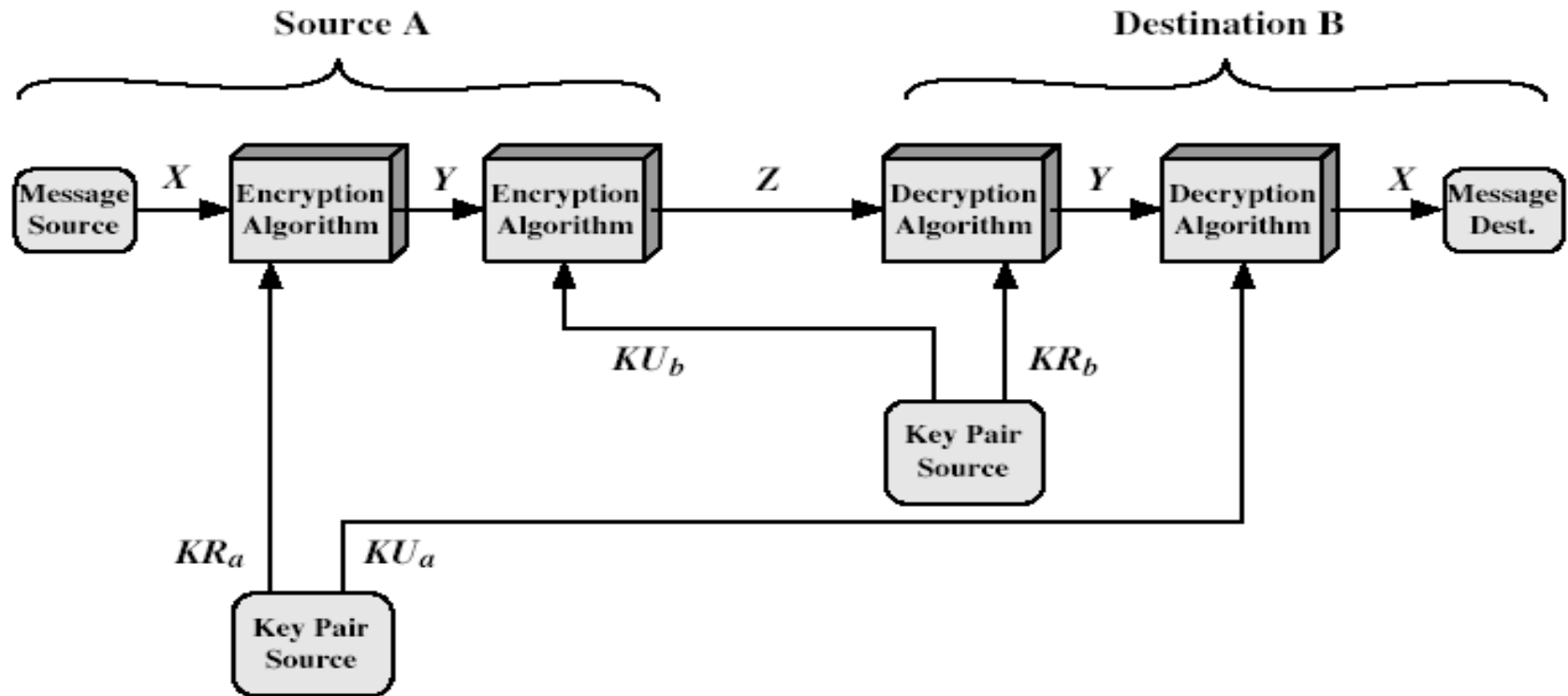


Figure 9.4 Public-Key Cryptosystem: Secrecy and Authentication

- Two major applications:
 - **encryption/decryption** (provide secrecy)
 - **digital signatures** (provide authentication)

Outlines

- Strength/weakness of DES, AES
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- RSA

Modular Arithmetic

- Public key algorithms are based on modular arithmetic.
- Modular addition.
- Modular multiplication.
- Modular exponentiation.

Modular Addition

- Addition modulo (mod) K
 - Poor cipher with $(d_k + d_m) \bmod K$, e.g., if $K=10$ and d_k is the key.

+	0	1	2	3	4	5	6	7	8	9
0	0	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9	0
2	2	3	4	5	6	7	8	9	0	1
3	3	4	5	6	7	8	9	0	1	2

- Additive inverse: addition mod K yields 0.
- "Decrypt" by adding inverse.

Modular Multiplication

- Multiplication modulo K

*	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9
2	0	2	4	6	8	0	2	4	6	8
3	0	3	6	9	2	5	8	1	4	7

- Multiplicative inverse: multiplication mod K yields 1
- Only some numbers have inverse

Modular Multiplication

- Only the numbers *relatively prime* to n will have mod n multiplicative inverse
- x, m relative prime: no other common factor than 1
 - Eg. 8 & 15 are relatively prime - factors of 8 are 1,2,4,8 and of 15 are 1,3,5,15 and 1 is the only common factor

Totient Function

- Totient function $\phi(n)$: number of integers less than n relatively prime to n
 - if n is prime,
 - $\phi(n)=n-1$
 - if $n=p*q$, and p, q are primes, $p \neq q$
 - $\phi(n)=(p-1)(q-1)$
 - E.g.,
 - $\phi(37) = 36$
 - $\phi(21) = (3-1) \times (7-1) = 2 \times 6 = 12$

Modular Exponentiation

x^y	0	1	2	3	4	5	6	7	8	9
0		0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1
2	1	2	4	8	6	2	4	8	6	2
3	1	3	9	7	1	3	9	7	1	3
4	1	4	6	4	6	4	6	4	6	4
5	1	5	5	5	5	5	5	5	5	5
6	1	6	6	6	6	6	6	6	6	6
7	1	7	9	3	1	7	9	3	1	7
8	1	8	4	2	6	8	4	2	6	8
9	1	9	1	9	1	9	1	9	1	9

Modular Exponentiation

- $x^y \bmod n = x^{y \bmod \phi(n)} \bmod n$
- if $y = 1 \bmod \phi(n)$ then $x^y \bmod n = x \bmod n$

Outlines

- Strength/weakness of DES, AES
- Public Key Cryptography
- Modular Arithmetic
- RSA

RSA (Rivest, Shamir, Adleman)

- The most popular one.
- Support both public key encryption and digital signature.
- Assumption/theoretical basis:
 - Factoring a big number is hard.
- Variable key length (usually 512 bits).
- Variable plaintext block size.
 - Plaintext must be "smaller" than the key.
 - Ciphertext block size is the same as the key length.

What Is RSA?

- To generate key pair:
 - Pick large primes (≥ 256 bits each) p and q
 - Let $n = p \cdot q$, keep your p and q to yourself!
 - For public key, choose e that is relatively prime to $\phi(n) = (p-1)(q-1)$, let $\text{pub} = \langle e, n \rangle$
 - For private key, find d that is the multiplicative inverse of $e \bmod \phi(n)$, i.e., $e \cdot d = 1 \bmod \phi(n)$, let $\text{priv} = \langle d, n \rangle$

RSA Example

1. Select primes: $p=17$ & $q=11$
2. Compute $n = pq = 17 \times 11 = 187$
3. Compute $\phi(n) = (p-1)(q-1) = 16 \times 10 = 160$
4. Select e : $\gcd(e, 160) = 1$; choose $e=7$
5. Determine d : $de=1 \pmod{160}$ and $d < 160$ Value is $d=23$ since $23 \times 7 = 161 = 10 \times 160 + 1$
6. Publish public key $KU = \{7, 187\}$
7. Keep secret private key $KR = \{23, 17, 11\}$

How Does RSA Work?

- Given $\text{pub} = \langle e, n \rangle$ and $\text{priv} = \langle d, n \rangle$
 - encryption: $c = m^e \bmod n, m < n$
 - decryption: $m = c^d \bmod n$
 - signature: $s = m^d \bmod n, m < n$
 - verification: $m = s^e \bmod n$
- given message $M = 88$ (nb. $88 < 187$)
- encryption:
$$C = 88^7 \bmod 187 = 11$$
- decryption:
$$M = 11^{23} \bmod 187 = 88$$

Why Does RSA Work?

- Given $\text{pub} = \langle e, n \rangle$ and $\text{priv} = \langle d, n \rangle$
 - $n = p * q, \phi(n) = (p-1)(q-1)$
 - $e * d = 1 \bmod \phi(n)$
 - $x^{e*d} = x \bmod n$
 - encryption: $c = m^e \bmod n$
 - decryption: $m = c^d \bmod n = m^{e*d} \bmod n = m \bmod n = m$
(since $m < n$)
 - digital signature (similar)

Is RSA Secure?

- Factoring 512-bit number is very hard!
- But if you can factor big number n then given public key $\langle e, n \rangle$, you can find d , hence the private key by:
 - Knowing factors p, q , such that, $n = p * q$
 - Then $\phi(n) = (p-1)(q-1)$
 - Then d such that $e * d = 1 \bmod \phi(n)$
- Threat
 - Moore's law
 - Refinement of factorizing algorithms
- For the near future, a key of 1024 or 2048 bits needed

Symmetric (DES) vs. Public Key (RSA)

- Exponentiation of RSA is expensive !
- AES and DES are much faster
 - 100 times faster in software
 - 1,000 to 10,000 times faster in hardware
- RSA often used in combination in AES and DES
 - Pass the *session key* with RSA